

Purification and Identification of an Unknown Solid

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1. EXPERIMENTAL METHOD

The $\text{CuCl}_2 \cdot 2\text{H}_2\text{O}$ (1.52 g) and KI (5.24 g) were combined in a 250 mL beaker for synthesizing the copper iodide compound. Two solids were stirred until distributed uniformly. Next, prepared 20 mL of deionized water was poured into the beaker. The mixture was stirred for 5 mins and then poured into a sucker filter with filtering paper. Another 300 mL of deionized water was used for washing (20 mL each time) the solid product until the color of the solid was consistent. Then the washed solid and the filtering paper were transferred to a crystallizing dish for drying. Finally, the solid was collected and weighed. Porcelain crucible was heated by Bunsen burner with blue part of the flame for 10 minutes. Next, the crucible was placed into a desiccator for cooling. The synthesized compound solid (0.3 ± 0.0001 g) were placed in the dried crucible weighted using the analytical balance.

Then, the crucible with the dried solid was heated for 30 minutes until it had a consistent color. After the reaction, the crucible with the compound was placed in the desiccator until it cooled to room temperature. CuO (0.1618 g) was obtained after the procedure conducted above.

2. RESULT

Table I. Actual yield, mass of the CuI_x solid placed into the crucible, and the mass of CuO

Actual Yield (<i>g</i>)	CuI_x (<i>g</i>)	CuO (<i>g</i>)
1.30	0.3000	0.1618

Table II. Theoretical percent copper in copper(I) iodide and copper(II) iodide

Theoretical percent of Cu in	CuI (%)	CuI_2 (%)
the compound	33.4	20.0

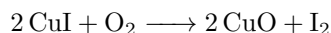
Table III. Experimental percent of copper in the synthesized copper-iodine compound, the theoretical yield of the compound, and the percent yield for the reaction

Cu in CuI (Exp. %)	The. Yield of CuI	Percent yield of reaction (%)
43.06	1.70	76.5

3. DISCUSSION

3.1. *Preparation and Determination of the Copper-iodine Compound*

The reaction formula and the possibilities of the types of compound (CuI_x $x=1$ or $x=2$) were provided.



The theoretical percent of copper in copper-iodine I and II were calculated to determine the value of x .

The following reasoning steps were applied to calculate the percent of Cu in the CuI in the reaction, derived by the product CuO:

CuO (in g) – CuO (in mol) – Cu (in mol) – Cu (in g) – Cu/CuI (in g) – Cu % – Comparison – Determination

Compared the experimental data (Table III 41.3 %) with the theoretical values (33.4 % table II), the $x = 1$ was selected since the value of CuI based on the experimental data and analysis was closer to the theoretical value of the CuI_2 (20.0 %).

Therefore, the compound was CuI.

3.2. *Uncertainty and Error*

As the relative low actual yield 76.5 % shown, the actual yield was significant lower than the theoretical yield. The certain amount Cu of was missed in the reaction or before the reaction. In the weight measurement of the CuI, the solid was wetted by the moisture of the air in the lab which caused the actual weight of CuI was lower than the measured value.